Coursework – Lab Report

Part 1 – Wax Composites Lab

1. Introduction

Wax-based composite materials are materials formed by adding fibers (fine wires, gauze, and wood chips) to wax-based materials in different combinations and arrangements, which often used in construction and casting. Compared with wax-based materials, wax-based composite materials show significant changes in mechanical strength [1] and anti-deformation ability.

In this experiment, candles were used as the matrix, red cotton thread, 0.7 mm carbon core and gauze were introduced as reinforcing materials to prepare six groups of wax-based composite samples. Due to the low strength and obvious brittleness of paraffin at room temperature, the simple bending test can directly reflect the role of the reinforcing materials in bearing, crack bridging and anti-fracture [2]. At the same time, the experiment changed the type, quantity and laying direction of the reinforcing materials to compare the influence of different reinforcing structures on the ultimate bending stress and failure mode, thereby analyzing the role of fiber reinforcement, interface bonding and two-dimensional continuous reinforcing layers in wax-based composite materials.

2. Experimental Methods

2.1 Material Selection and Experimental Design

All samples were made using the same batch of candles as the base material to reduce the influence of differences in wax matrix components. Candle wax is prone to brittle fracture at room temperature, so whether continuous fibers can bear tensile loads, bridge cracks and improve toughness can be clearly demonstrated through simple bending tests.

Table 2-1-1. Main Materials, Tools and Their Functions in This Experiment

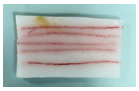
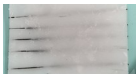
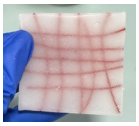
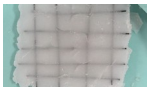
Category	Material/Tool	Role in Experiment	Selection Reason or Control Point
Matrix	Candle wax	Forming a wax-based composite material matrix	Used from the same batch to reduce differences in matrix composition
Flexible reinforcing agent	Red cotton thread	0° unidirectional or 0°/90° grid reinforcement	Continuous, easily directional, capable of demonstrating fiber bridging and crack resistance effect
Rigid reinforcing body	0.7 mm automatic pencil lead/carbon core	Unidirectional or grid-like rigid reinforcement	Compare with cotton thread, analyze the influence of the stiffness and brittleness of the reinforcing body on the failure mode
Two-dimensional reinforcement layer	Gauze	Forming a continuous fabric reinforcement layer	Used to observe the influence of the two-dimensional continuous layer on crack propagation and toughness

Forming tool	2 student cards + 2 rulers	Assemble to form a rectangular pouring area and control the boundaries	It is more conducive to limiting the deformation caused by hot wax compared to elastic plastic molds; the rulers can directly measure the thickness.
Demolding aid	Oiled paper	Reduce the adhesion between the wax-like substance and the bottom surface	Facilitating demolding and avoiding the wrinkling of the oil paper which could cause uneven edges

2.2 Sample Design and Control Relationship

The sample design adopts a single-variable control approach, with each sample forming a direct comparison relationship with at least one other sample. To distinguish the effects of factors such as fiber type, fiber quantity, fiber orientation, mold geometry, and two-dimensional continuous reinforcement layers, the specific designs of the six groups of samples are as follows:

Table 2-2-1. The reinforcement bodies, architectures, and comparison purposes of wax-based composite materials samples.

No.	Photo	Name	Enhancer and Framework	Comparison Objective
S1		Red-5-Uni	5 cotton threads; 0° unidirectional rectangular plate	Cotton thread unidirectional reinforcement reference group
S2		Graphite-5-Uni	5 0.7 mm lead cores; 0° unidirectional	Difference in flexibility of fibrous material and rigid brittle reinforcing body compared with R5-U
S3		Red-10-Mesh	5 + 5 cotton threads; 0° / 90° grid	Cross-laid and transverse fiber effects compared with R5-U
S4		Red-10-Mesh-Fabric	Cotton thread grid + 1 layer of gauze	Evaluation of the influence of two-dimensional continuous fabric on crack propagation and toughness
S5		Graphite-10-Uni	10 lead cores; 0° unidirectional, high reinforcement content	Effect of reinforcing body quantity/volume fraction compared with G5-M+F
S6		Graphite-10-Mesh	5 + 5 lead cores; 0° / 90° grid	Architecture effect of rigid reinforcing body compared with G5-M+F /G10-U

S1–S2 compare reinforcement type; S2–S5 compare content; S1–S3 and S5–S6 compare architecture; S3–S4 compare gauze effect.

2.3 Preparation Process

The size of elastic plastic bowl-shaped molds was relatively large, and it tended to bulge and deform locally when in contact with the hot wax liquid, making it difficult to stabilize the sample thickness. Moreover, the secondary moment of the rectangular cross-section in the bending process is related to the thickness in a cubic relationship ($I = bh^3/12$), and the difference in thickness would significantly affect the subsequent bending test results. Therefore, this group did not adopt the elastic plastic

mold as the final molding solution. Instead, we used the "student card + ruler" temporary rigid boundary method to prepare the samples.

Specifically, as shown in Figure 2-3-1, a layer of the built-in oil paper is first laid flat as the bottom of the mold to facilitate demolding; then, two student cards are used as the front and back partitions, and two rulers are used as the left and right side walls to assemble and form the rectangular pouring area in the middle. The rulers are placed vertically so that the side markings on their surfaces can be directly read during the pouring process, thereby minimizing the variation in the target thickness of each sample. Although this method still belongs to a simple manual mold, compared to the elastic plastic molds that are prone to deformation when heated, it is more suitable for the requirement of consistent thickness in this experiment.

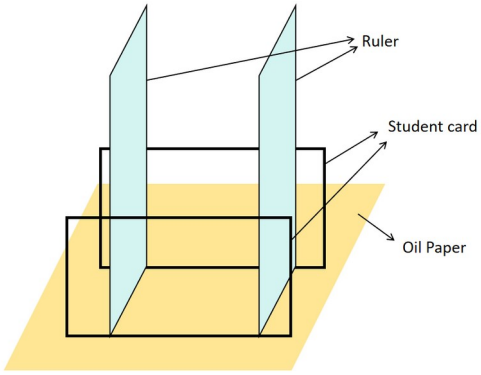


Figure 2-3-1. Schematic diagram of the improved wax-based composite material pouring mold: The bottom is a layer of oil paper, two student cards and two rulers together form a rectangular pouring area. The scale on the side of the ruler is used to control the thickness of the wax liquid during pouring.

The specific sample preparation process is as follows:

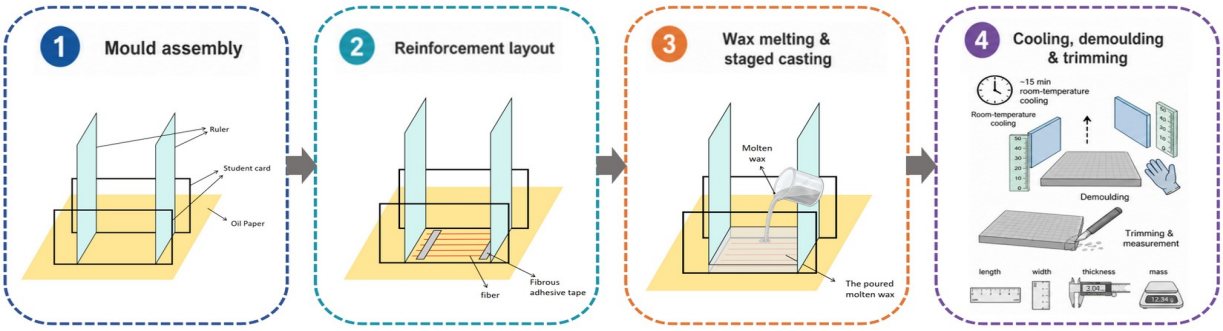


Figure 2-3-2 Flowchart of sample preparation for wax-based composite materials.

Mould assembly: Place a layer of the oil paper brought by the group as the bottom of the mold, and place two student cards and two rulers perpendicular to the tabletop, forming a rectangular area in the middle, while trying to make their edges closely contact each other to reduce wax leakage and irregularities at the product edges. Also, pay attention to that the scale of the ruler should face outward for later reading and control of the height of the wax liquid.

Reinforcement layout: According to the sample design, cut the red cotton thread, 0.7mm carbon core and gauze in advance. During sample preparation, arrange them in the rectangular area as required by the sample and fix them on the oil paper with tape to prevent fibers from drifting during the pouring process. For cotton thread samples, keep the fiber direction

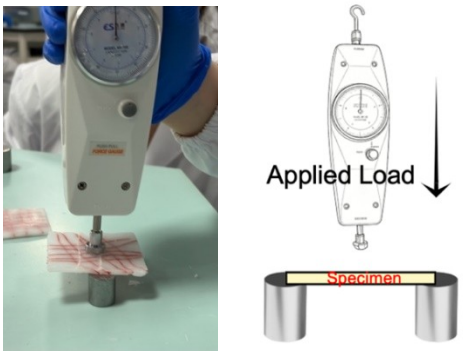
straight and slightly tightened; for carbon core samples, try to make them parallel and have uniform spacing; for 0°/90° samples, first lay the first direction support structure, then lay the second direction support structure vertically to form a grid; for samples with gauze, ensure the gauze is flat and covers the main stress area.

Wax melting & staged casting: Cut the wax into small pieces and heat them until fully melted. During pouring, add a small amount of wax liquid first, allowing it to penetrate into the fiber gaps and just cover the fibers, wait for about 1 minute, when the viscosity of the wax liquid slightly increases, fix the position of the support structure, then remove the tape on the side, add more wax liquid to the target thickness. Also, during the pouring process, observe whether there are fiber drifts, residual bubbles or local volume accumulation phenomena.

Cooling, demolding & trimming: To reduce the risk of thermal shrinkage cracks, the samples are cooled naturally at room temperature for about 15 minutes. No water cooling or forced air cooling is used. After complete curing, remove the student cards and rulers, peel off the samples. Then trim the edges evenly and record the length, width, thickness and weight of the sample.

2.4 Testing procedure

All wax-based composite samples were compared using the same improvised three-point bending setup. The samples were placed horizontally on two cylindrical supports with a fixed span of 4.00 cm; then, a handheld force gauge was used to gradually apply a vertical load at the mid-span of the samples and record the video at the same time. During the test, the deflection, crack initiation, crack propagation, and final fracture behavior of the samples were continuously observed. When the samples fractured or lost structural integrity, the reading at that moment was recorded as the failure load, and the fracture surface and macroscopic failure morphology were photographed. For samples that did not immediately fracture but showed obvious cracks, the crack initiation position, fiber pull-out/breakage phenomena, and residual deformation were also recorded to explain the fiber bridging, matrix-fiber bonding, and failure mode in the discussion.



(a) (b)

Figure 2-4-1. Three-point bending test configuration for the wax-composite specimens: (a) experimental photograph of the improvised setup; (b) schematic diagram showing the specimen placed on two supports with a span of 4.00 cm and subjected to a centrally applied load.

3. Results

Table 3-1. Origin data of sample and fracture load

Sample Name	Mass *m* (g)	Length *a* (cm)	Width *b* (cm)	Thickness *h* (cm)	Fracture Load *P* (N)
R10-M	10.56	6.20	5.91	0.30	11.23
G5-U	10.38	6.79	4.21	0.35	13.30
G10-M	8.93	5.75	4.02	0.31	15.32
G10-U	9.69	6.52	2.45	0.46	16.06
R5-U	10.41	5.75	5.35	0.33	20.30
R10-M+F	10.57	5.73	5.33	0.38	30.80

In terms of fracture load, the R10-M sustained the smallest load, approximately 11.23 N, whereas the R10-M+F withstood the largest load, at 30.8 N.

Assuming the specimens are rectangular prisms. Because specimen dimensions differed, fracture load was normalised to ultimate flexural stress using the three-point bending equation

$$\sigma_f = \frac{3FL}{2bh^2}$$

Here, F is the applied force at fracture, and L is the distance between the two supports, which was uniformly set at 4 cm in this experiment. The results are presented in the table below.

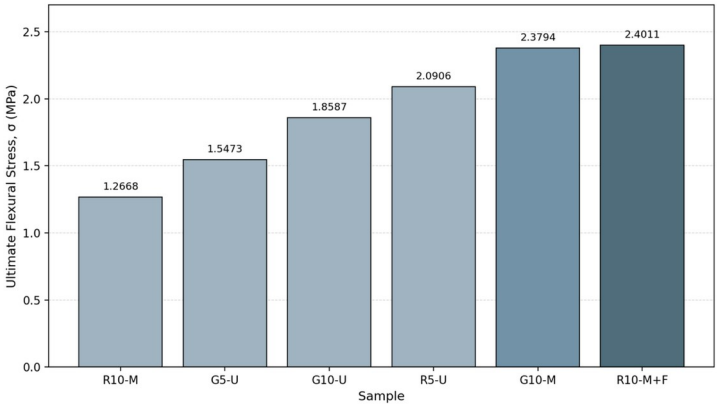


Figure 3-1. Comparison of the ultimate flexural stress of six wax-composite specimens measured by three-point bending. R10-M+F showed the highest flexural stress, while R10-M showed the lowest.

4. Discussion

4.1 Flexural Load-Bearing Performance

The flexural properties of wax-based composites are influenced by the reinforcing material, fiber content, reinforcement structure, and specimen geometry. Because the dimensions of each specimen vary, the fracture load alone is insufficient for comparison. Therefore, the ultimate flexural stress is a more reliable indicator. The properties increase in the following order:

$$R10\text{-}M < G5\text{-}U < G10\text{-}U < R5\text{-}U < G10\text{-}M < R10\text{-}M+F$$

The effectiveness of reinforcing elements depends on their quantity and their ability to bear loads in the flexural-tensile zone. Cracks typically initiate first on the tension side of the specimen. If the reinforcing phase can effectively share the tensile load, it can increase the bending stress the specimen can withstand before macroscopic fracture occurs.

For graphite-core-reinforced specimens, G10-U has a higher ultimate bending stress than G5-U, indicating that increasing the number of reinforcing elements can improve load-bearing capacity to

some extent. However, this improvement was not significant, suggesting that the interfacial bonding between the graphite cores and the wax matrix may be weak, making it difficult to achieve efficient load transfer with the wax matrix during bending. Therefore, although increasing the number of graphite cores improves local stiffness, the reinforcement efficiency is limited by interfacial bonding and brittle fracture. The bending stress of G10-M is higher than that of G5-U and G10-U, indicating for a stiffening element, a 0°/90° grid structure is more effective than a unidirectional arrangement in dissipating local stresses. The grid structure restricts crack propagation in both directions and forms a more continuous support network in the loading zone, thereby exhibiting a higher ultimate bending stress.

R10-M+F exhibited the highest ultimate bending stress of 2.4011 MPa. Compared to discrete red fibers, the gauze layer provides a more continuous two-dimensional reinforcement network capable of distributing loads over a larger area and enhancing the overall structural integrity of the specimen. Consequently, the addition of gauze not only improves the composite's peak load-bearing capacity but also maintains good overall connectivity even after crack initiation. It should be noted that the difference in ultimate bending stress between R10-M+F and G10-M is relatively small, indicating that the two are similar in peak strength; however, there are significant differences in their post-fracture integrity and failure mechanisms, which will be further discussed in the failure mode section.

Since all the samples were prepared manually, the thickness, fiber orientation, reinforcement spacing, internal bubbles and local defects were difficult to be fully controlled. Among them, the ultimate bending stress of the R5-U sample was abnormally high, which might be related to the measurement error of the sample size, especially the thickness measurement error. The bending stress in the three-point bending formula is very sensitive to the thickness, so even a small deviation in thickness could also cause significant calculation errors.

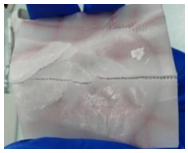
A major limitation is that no unreinforced wax control specimen was tested. Therefore, the results demonstrate relative differences among reinforcement architectures, but do not quantify the absolute improvement over neat wax.

4.2 Failure Modes and Fracture Mechanisms

Three distinct fracture failure modes were observed across the six specimens, summarized in Table 4-2. The classification is based on the post-fracture morphology of each specimen and the relationship between the matrix crack and the reinforcement at the fracture surface.

Table 4-2. Observed fracture failure modes of the wax-based composite specimens, grouped by reinforcement type and architecture.

Name	Sample Photo	Fracture failure mode	Detail
G5-U		Co-fracture (fibre and matrix break together)	The wax matrix and the graphite cores fractured together in essentially the same plane. The lead cores snapped along with the matrix, leaving a clean, near-monolithic fracture face. Failure occurred suddenly with little prior visible deformation, and the two halves of the specimen separated completely.
G10-U			
G10-M			

R5-U R10-M	 	Intact-fibre crack bridging	The wax matrix cracked transversely on the tensile side, but the cotton threads remained intact across the open crack and held the two halves of the specimen together. The fibres stretched across the crack mouth without snapping, and no clean separation of the specimen was observed even after matrix fracture.
R10-M+F		Fabric crack-arrest (continuous 2-D bridging)	The wax matrix developed cracks during loading but the specimen did not separate into two pieces. The continuous gauze layer remained coherent across the cracked region and held the cross-section together as a sheet.

The three observed modes can be traced back to the interaction between the strain-to-failure of the reinforcement and that of the brittle wax matrix. Candles wax has a low tensile strain-to-failure, so the strain at which the matrix cracks effectively sets the strain at which any reinforcement is challenged. Pencil-lead graphite is itself brittle, with a smooth, chemically inert surface that develops only weak adhesion with paraffin wax; its strain-to-failure is comparable to that of the matrix and its interfacial bond is poor. When the matrix crack reaches a graphite core, the core therefore breaks in the same plane rather than deflecting the crack or pulling out, and the composite fails in a near-monolithic brittle manner (G5-U, G10-U, G10-M). Cotton thread, by contrast, has a much higher strain-to-failure and a rough, twisted surface that promotes mechanical interlocking with the matrix. The cotton fibres consequently remain intact across the open matrix crack and continue to carry tensile load — the classical intact-fibre crack-bridging mechanism (R5-U, R10-M).

Whether bridging actually arrests crack propagation, however, depends on the reinforcement architecture. In R5-U and R10-M, bridging only acts at the discrete points where individual threads cross the fracture plane, and the crack can propagate freely through the large unreinforced regions of brittle wax between them. In R10-M+F, the gauze intersects the fracture plane along its entire length, so bridging is effectively continuous across the cross-section. The matrix still cracks, but the specimen does not separate; energy is additionally dissipated through fibre stretching, frictional pull-out at the yarn interfaces, and progressive matrix–fibre debonding. This is consistent with R10-M+F being the only sample to remain coherent after matrix fracture.

5. Conclusion

This experiment used ultimate bending stress as the primary evaluation index to investigate the effects of different reinforcement types, contents and arrangements on the three-point bending performance of wax-based composites. The results show that the flexural performance of the composites was strongly influenced by the structural form and distribution of the reinforcing materials.

The R10-M+F exhibited an ultimate flexural stress of 2.4011 MPa, the highest of all test results, indicating the greatest improvement in flexural performance. It can therefore be concluded that the combination of a continuous two-dimensional reinforcement layer with a grid-like structure helps to disperse local stresses more effectively and delays crack propagation during loading. In contrast, specimens with a lower reinforcement content or a simpler reinforcement structure exhibited faster crack growth and a more abrupt failure process.

Further optimization of the reinforcement layout, number of layers and matrix–reinforcement bonding may improve the mechanical response and overall performance of this type of composite material.

6. References

[1] Hutton-Prager, B., Adenekan, K., Sypnewski, M., Smith, A., Meadows, M., & Calicdan, C. (2021b). Hydrophobic development and mechanical properties of cellulose substrates supercritically impregnated with food-grade waxes. Cellulose, 28(3), 1633–1646. Available from: https://doi.org/10.1007/s10570-020-03628-2
[2] Dr. Shafique Ahmed Wax and Layup Lab Handouts, Available from:https://qmplus.qmul.ac.uk/course/view.php?id=28385

Part 2 – Composite Layup for Aircraft Stowage Panel

1. Introduction

Above the seats of civilian aircraft, there are usually racks for storing luggage. To ensure that the luggage remains safely stored during turbulence, these racks are usually required to be able to withstand a weight of 25 kilograms. Therefore, these racks place great emphasis on the selection of materials. Currently, glass fiber / epoxy resin laminates have been widely used in aviation components due to their excellent comprehensive performance [1]. This material can adjust the fiber layup direction and stacking sequence to cope with different stress conditions, thereby enhancing the structural load-bearing capacity.

In this experiment, using glass fiber and epoxy resin as raw materials, multi-layer composite plates were prepared by manual layup process [2]. Three types of manually laid laminated plates were prepared, two-layer parallel laminated plate, two-layer laminated plate with an angle of 0° and 45°, and a four-layer laminated plate with angles of 0°, 45°, -45° and 0°. Then the deflection of each sample was tested through bending experiments and the sample's EI was calculated, thereby determining whether the sample could bear a load of 25kg and analyzing how different layup designs affect the sample's bending stiffness. At the same time, a uniaxial tensile test was also conducted and compared with the EI obtained from the bending test.

2. Experimental Methods

2.1 Layering Strategy and Design Basis

The engineering scenario corresponding to this experiment is the luggage rack panel above the aircraft. The design objective is not merely to prepare the "thickest" or "hardest" sheet, but to ensure that the glass fiber/epoxy laminated board has sufficient bending stiffness and structural integrity under the conditions of limited material usage and manual layering. The static load of 25 kg luggage is approximately 245 N, so the subsequent tests focus on the bending angle, deflection, and equivalent bending stiffness EI of the panel under central loading.

Considering that the aircraft interior components are also limited by weight, material cost, and manufacturing complexity, this group sets up two two-layer low-material-usage control groups and one four-layer reinforced design group. This allows for comparing the influence of fiber orientation on performance and determining whether the increase in the number of layers leads to sufficient stiffness improvement to offset the additional consumption of fiberglass fabric, resin, and preparation time.

Table 2-1-1 Lamination schemes and design effects of glass fiber/epoxy composite panel.

Panel	Layer configuration	Number of layers	Design function
Two-layer parallel	[0°/0°]	2	The main direction bearing reference

			group with low material usage
Two-layer diagonal	[0°/45°]	2	Evaluate the oblique fiber/shear response without increasing the number of layers
Four-layer hybrid	[0°/+45°/−45°/0°]	4	The multi-directional reinforcement design group, evaluate the stiffness gain after adding materials

The two-layer parallel laminate [0°/0°] is the main stiffness reference benchmark with the lowest material consumption. The orientations of the two layers of glass fiber cloth are completely consistent, and the 0° fibers mainly bear the tensile and compressive stresses on the outer surface of the specimen when it undergoes bending, thus it is expected to have higher main-direction stiffness in the three-point bending test. Its drawback is obvious anisotropy, and its adaptability to oblique loads, shear deformation, and local non-uniform loading is limited.

The two-layer diagonal laminate serves as the second low-material-consumption control group. Its purpose is not simply to maximize the main cross-direction stiffness, but to improve the response of off-axis deformation and shear-dominated deformation by changing the fibre orientation without increasing the number of glass fiber layers and resin consumption. The comparison with the [0°/0°] plate can illustrate the engineering concept of "first optimizing the layup, then increasing the material".

The four-layer hybrid laminate [0°/+45°/−45°/0°] is the enhancement scheme of this experiment. The outer 0° layer is far from the neutral axis and contributes more to the EI in bending; the middle +45° and −45° layers are used to improve the shear response, limit the rapid expansion of cracks along a single direction, and improve the stability of the laminated plate under multi-directional loading. This scheme is expected to have the highest performance, but it also has higher quality, resin consumption, and preparation complexity, so the final evaluation must be made by jointly considering the stiffness improvement and material increase.

Table 2-1-2 The mechanical effects of different fibre orientations in the layup.

Fiber orientation	Role in the laminate	Design rationale
0°	Main direction bending bearing	Bear the main tensile/compressive stress along the length direction of the sample
+45°	Shear response and crack deflection	Improve the adaptability to non-axisial loads
−45°	Form a balanced inclined layer with +45°	Reduce the tendency of unilateral oblique deformation and enhance the multi-directional stability
Lateral 0° layer	Increase the bending stiffness	The outer layer is farther from the neutral axis, contributing more to EI

Therefore, the layup strategy of this experiment is essentially a comparison of performance and material efficiency. Subsequently, the elastic modulus (EI) is calculated based on the load-deflection data, and the number of layers, thickness, and sample size are combined to determine which layup is more reasonable in terms of bearing capacity, material consumption, lightweighting, and manufacturing feasibility.

2.2 Fabrication documentation

2.2.1 Materials, Equipment and Control Variables

All three composite material panels were fabricated using manual wet layup method to ensure that the experiment could meet the requirements of the course, namely "manual hand layup" and "improvised testing". To enhance the comparability between different layups, this group tried to keep the resin/curing agent ratio, room temperature curing conditions, manual brushing method, curing time and subsequent test reading methods consistent; the main variables focused on fiber orientation and layer number. It should be noted that due to the absence of vacuum bag compaction or rigid mold constraints, there may still be deviations in resin content, thickness and local pores. These factors need to be discussed as error sources in the interpretation of the subsequent results.

- Matrix: epoxy resin and hardener, mixed at a 1:1 ratio by volume.
- Reinforcement: woven glass fibre cloth.
- Layup substrate: standard kitchen aluminum foil, used as both working surface and intended release liner.
- Cutting tools: scissors, ruler, marker, and paper tape (used to stabilise the weave during cutting).
- Resin handling: mixing containers, stirring sticks, and brushes for resin application.
- PPE: nitrile gloves, safety glasses, and lab coat.

2.2.2 Cutting procedure

To produce fibre plies with clean edges and prevent fraying of the woven cloth, paper tape was applied across a rectangular region marked on the glass fibre cloth prior to cutting. Scissors were then used to cut along the taped region, with the adhesive tape stabilising the weave and constraining individual fibres during cutting (Figure 2-2-1). The residual tape along the cut edges was intentionally retained on the finished plies, as attempting to remove it would cause the edge fibres to unravel and compromise the ply geometry. The retained tape therefore remained embedded in the panel after curing and is visible along the ply edges in the cured samples.

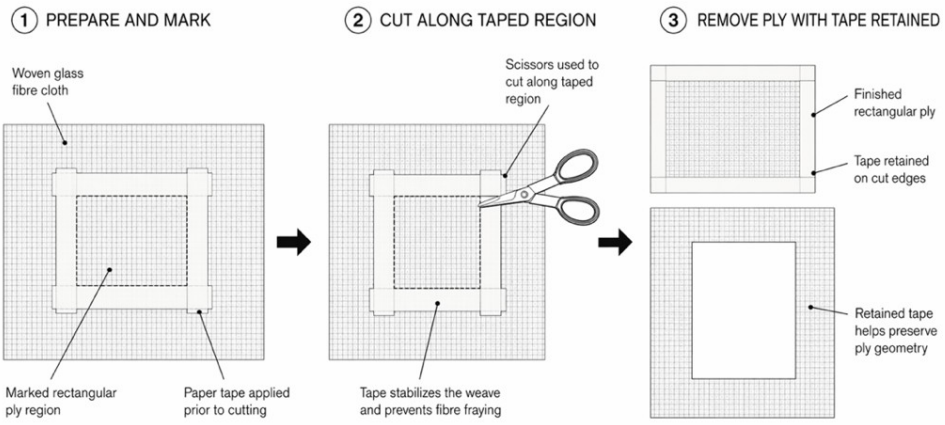


Figure 2-2-1. Cutting procedure for the glass fibre plies. Paper tape was applied over the marked cutting line prior to cutting to stabilize the weave and prevent edge fibres from fraying.

2.2.3 Layup procedure

Each panel was fabricated by hand wet layup on an aluminum foil substrate, which served as both a working surface and a flexible liner. The layup procedure is illustrated schematically in Figure 2-2-2 and was carried out as follows:

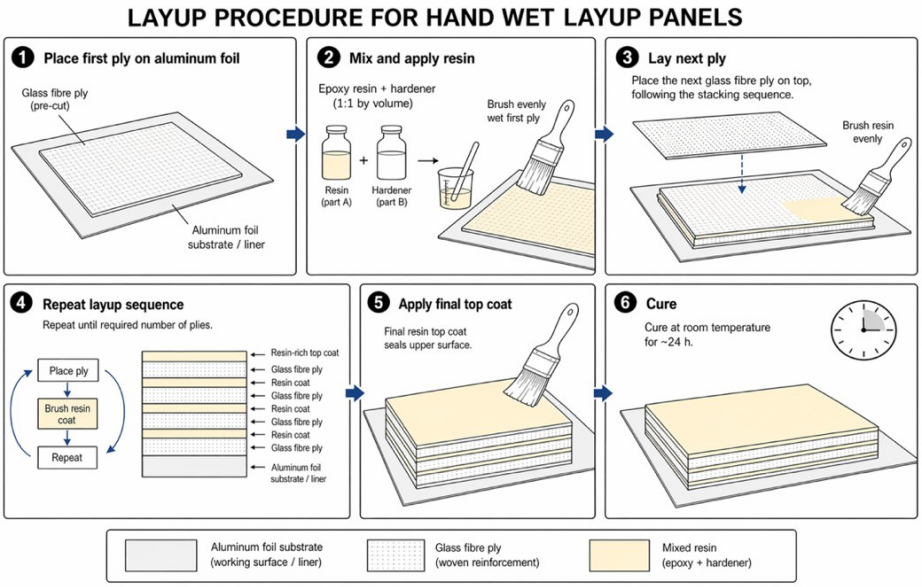


Figure 2-2-2. Schematic of the hand wet layup procedure: a fibre ply is placed on the aluminum foil substrate, resin is brushed over it, and the next ply is laid on top before repeating the sequence. A final top coat of resin seals the upper surface.

- A pre-cut glass fibre ply was placed flat onto the aluminum foil.
- The epoxy resin and hardener were freshly mixed at a 1:1 ratio by volume, and the mixed resin was brushed evenly across the surface of the ply until the weave was fully wetted.
- The next ply was laid directly on top, oriented according to the panel's stacking sequence (Table 2-1-1), and a further coat of mixed resin was brushed across its surface.
- The previous step was repeated until the required number of plies had been placed (two for the two-layer panels, four for the hybrid panel).
- After the final ply had been resin-coated, a top coat of resin was applied to seal the upper surface.
- The completed layup was left undisturbed to cure at room temperature for approximately 24 hours.

2.2.4 Observations of cured panels

Once cured, the panels were visually inspected prior to testing. Several defects and non-idealities were identified, which were common to all three panels and are attributable to the improvised hand-layup route and the absence of a confining mould or vacuum consolidation.

- Fine air bubbles were distributed throughout the resin matrix (Figure 2-2-3a), visible as a dense speckled texture when viewed in transmission. These bubbles were presumed to originate from air entrained during manual brushing that was not released before the resin gelled.
- The resin distribution was not uniform across the panel area, with a visibly greater resin thickness toward the central region and a reduced resin content near the edges (Figure 2-2-3b). This is consistent with uncontrolled gravity flow of the unconstrained resin during the early stages of cure.
- The two major surfaces of each panel were asymmetric (Figure 2-2-3c): the bottom surface retained the aluminum foil substrate, which had bonded strongly to the cured resin and could not be peeled off, while the top surface was smooth, glossy, and essentially free of surface texture.
- Residual paper tape along the ply edges remained embedded in the cured resin and could not be removed post-cure without damaging the edge fibres (Figure 2-2-3d).

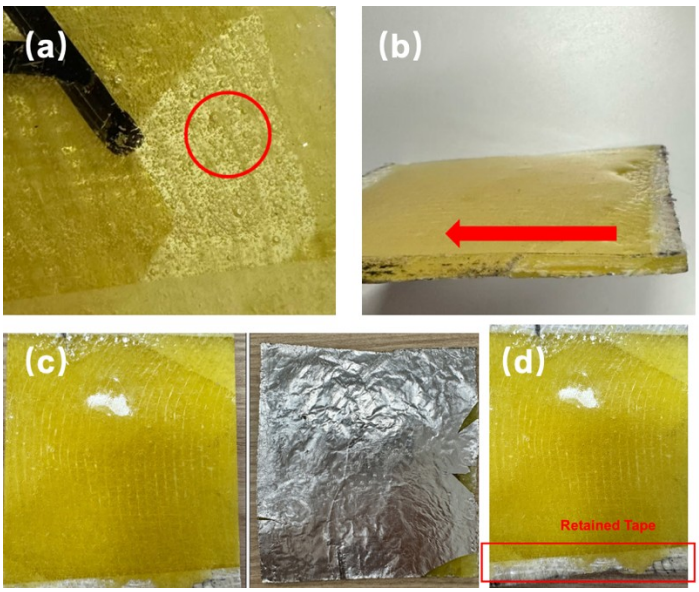


Figure 2-2-3. Defects observed in the cured composite panels: (a) fine air bubbles distributed through the resin matrix (highlighted by the red circle); (b) non-uniform resin thickness visible in cross-section, with the red arrow indicating the gradient from the thinner edge to the thicker central region. (c) asymmetric surfaces of a cured panel, showing the smooth glossy resin-rich top surface (left) and the aluminum foil substrate bonded to the underside (right); (d) residual paper tape retained along the ply edge (outlined in red) that could not be removed post-cure without damaging the edge fibres.

To reduce the influence of these defects on the test results, each cured panel was trimmed so that only the central region, in which the resin thickness was visually most uniform, was retained for subsequent testing. The four-layer hybrid panel was reserved for the bending test, while the two two-layer panels were used for both the bending test and, after further sectioning into strips, the tensile test. The final trimmed dimensions of the three panels used in the bending test are listed in Table 2-2-1.

Table 2-2-1. Trimmed dimensions of the three composite panels used in the bending test.

Panel	Length (cm)	Width (cm)	Thickness (cm)
Two-layer parallel [0°/0°]	11.20	7.95	0.30
Two-layer diagonal (45° between plies)	7.39	4.56	0.40
Four-layer hybrid [0°/+45°/−45°/0°]	8.88	7.71	0.45

2.3 Testing method

Two types of mechanical test were performed on the composite panels. A three-point bending test was carried out on all three panels using the same improvised setup employed in Part 1, in order to back-calculate their flexural rigidity and assess their suitability as load-bearing aircraft stowage panels. In addition, a tensile test was performed on strips cut from the two two-layer panels to characterize their in-plane tensile response. The four-layer hybrid panel was not tensile tested.

2.3.1 Bending test method

An improvised three-point bending test was used to assess the flexural behaviour of the composite panels. Each panel was placed horizontally on two supports to form a simple span, and a central load was applied at the mid-span using a handheld force gauge (Figure 2-3-1). As the three panels did not share identical plan dimensions (see Table 2-2-1), the support span was selected individually

for each panel according to its length and was recorded together with the corresponding dimensions for subsequent analysis.

The applied load was increased in incremental steps of 10 N (0, 10, 20, 30 N and so on), read directly from the force gauge, and the corresponding mid-span deflection was measured against an adjacent ruler at each loading step. Loading was deliberately terminated before the onset of visible damage or failure, so that the panels could be retained intact for subsequent evaluation. No cracking, delamination, or permanent deformation was observed in any panel within the loading range applied. Although the bending response is expressed in terms of the bending angle in the experimental manual, in this experiment, the mid-span deflection measured directly is used as the main deformation parameter. Under the condition of small deflection three-point bending, the bending angle and the mid-span deflection are two related measurement values of the same bending deformation. Therefore, using the standard simply supported beam model (with a central point load) and these load-deflection data, the flexural rigidity EI of each layer of composite material can be calculated inversely. Finally, the calculated stiffness values are compared with the target stiffness required to support a static load of 25 kilograms.

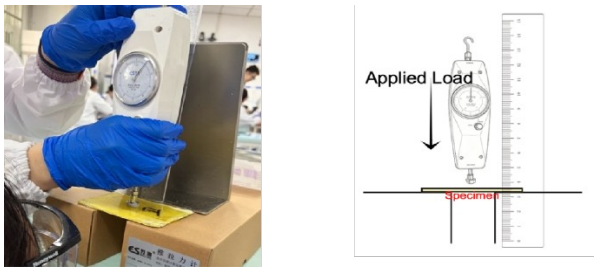


Figure 2-3-1. Bending test setup used for the composite panels: (a) photograph of the improvised experimental arrangement using a handheld force gauge; (b) schematic illustration of the loading configuration, showing the specimen subjected to a central applied load and the corresponding mid-span deflection measurement.

2.3.2 Stretching test method

A uniaxial tensile test was carried out on strips cut from the two two-layer panels in order to characterize their in-plane tensile response along the principal ply direction. The four-layer hybrid panel was not tensile tested.

One long rectangular strip was sectioned from each of the two-layer panels, with the long axis of the strip aligned with one of the 0° fibre directions in that panel. The strips were cut from the central region of each cured panel so that the test specimens came from the zone of most uniform resin thickness, as identified during post-cure inspection (see Figure 2-2-3). The strip dimensions are listed in Table 2-3-1.

Table 2-3-1. Dimensions of the tensile test strips cut from the two two-layer panels.

Source panel	Length (cm)	Width (cm)	Thickness (cm)
Two-layer parallel [0°/0°]	11.70	2.95	0.30
Two-layer diagonal (45° between plies)	7.15	2.75	0.31

The tensile tests themselves were carried out by the module instructor on a university tensile testing machine in a separate session; the two strips were handed over for testing and the corresponding data files were returned to the group.

3. Results

3.1 Experimental Results

3.1.1 Bending load-deflection results

Three-point bending tests were conducted on each specimen under increasing applied loads. The mid-span deflection was calculated from the change in ruler reading, as shown in Table 3-1-1. These load-deflection data provide the experimental basis for the EI back-calculation in Section 3.2.

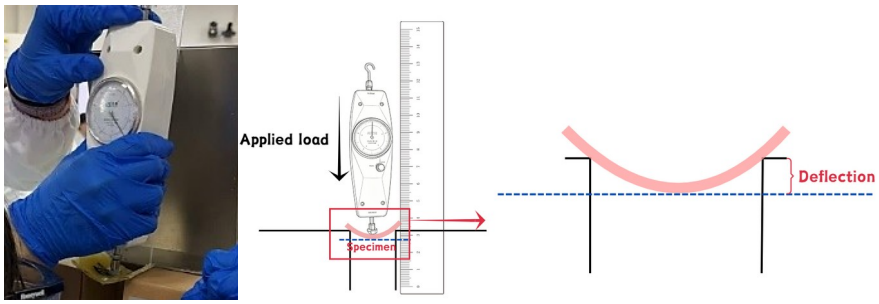


Figure 3-1-1. Measurement of the sample's deflection distance

Table 3-1. Applied load, measured mid-span deflection and calculated bending angle for each layup.

	Load /N	**Ruler reading / cm**	**Change in ruler reading / cm**	**Support span, L / cm**	**Mid-span deflection, δ / cm**	Angle / rad
Two-layer parallel [0°/0°]	0	7.61	0.00	5	0.00	0.0000
	10	7.50	0.11		0.11	0.0550
	20	7.40	0.21		0.21	0.1045
	30	7.05	0.56		0.56	0.2730
	40	6.40	1.21		1.21	0.5440
Two-layer diagonal (45° between plies)	0	7.61	0.00	8	0.00	0.0000
	10	7.21	0.40		0.40	0.0997
	20	6.91	0.70		0.70	0.1733
	30	6.15	1.46		1.46	0.3499
	40	5.35	2.26		2.26	0.5143
Four-layer hybrid [0°/+45°/−45°/0°]	0	7.61	0.00	4	0.00	0.0000
	10	7.61	0.00		0.00	0.0000
	20	7.60	0.01		0.01	0.0051
	30	7.60	0.01		0.01	0.0051
	40	7.60	0.01		0.01	0.0051
	50	7.59	0.02		0.02	0.0099
	60	7.58	0.03		0.03	0.0150
	70	7.58	0.03		0.03	0.0150
	80	7.55	0.06		0.06	0.0300
	90	7.53	0.08		0.09	0.0400
	100	7.50	0.11		0.11	0.0550
	110	7.49	0.12		0.12	0.0599
	120	7.40	0.21		0.21	0.1045

3.1.2 Tensile test results

The tensile tests provided an additional set of in-plane stiffness results for comparison with the bending-derived stiffness. As shown in Figure 3-3, the tensile moduli of the two-layer parallel [0°/0°] and two-layer angled laminates are 379.19 MPa and 240.02 MPa, with yield strains of 0.036 and 0.047, respectively.

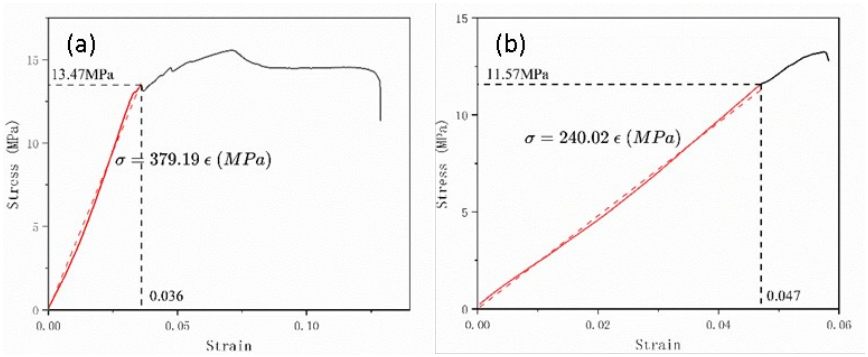


Figure 3-1-2. Tensile stress–strain curves of two specimens. (a) Two-layer parallel [0°/0°] layup (b) Two-layer angled layup

3.2 Three-point bending analysis

3.2.1 Back-calculation of flexural rigidity EI

For a three-point bending test, the maximum deflection at the center is given by

$$w_{max} = \frac{P\,L^3}{48\,EI}$$

where P is the applied load, L is the span length, E is the elastic modulus of the material, and I is the second moment of area. For a rectangular cross-section,

$$I = \frac{b\,h^3}{12}$$

The slope of the load-deflection curve, $k = w_{max}/P$, was obtained from linear fitting. Therefore, the flexural rigidity and equivalent modulus were calculated as:

$$EI = \frac{L^3}{48\,k} \quad \text{and} \quad E = \frac{L^3}{48\,kI}$$

The graph obtained from the three-point test is shown in the figure below.

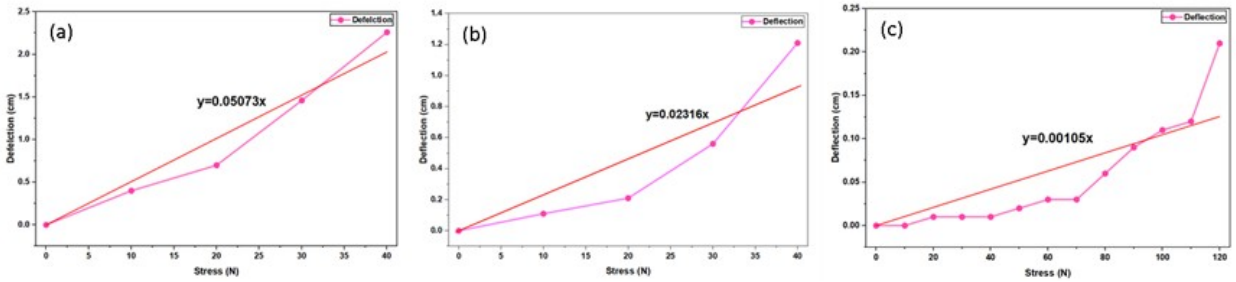


Figure 3-2-1. Load–deflection curves from three-point bending tests for three specimens. (a) Two-layer parallel (b) Two-layer angled (c) Four-layer

The calculation results of the equivalent elastic modulus and flexural rigidity are shown in Table 3-2. EI is the main structural stiffness index because it represents the bending resistance of the whole specimen. E is retained as an equivalent material-level index for comparison between different layup configurations.

Table 3-2-1 Back-calculated flexural rigidity and equivalent flexural modulus of each layup configuration.

Lay-up configuration	E (MPa)	EI (N·m^2)
Two-layer angled	45.5656	0.011082
Two-layer parallel	117.617	0.021039
Four-layer	227.734	0.133333

The EI values are approximate stiffness estimates because some two-layer specimens reached relatively large bending angles at higher loads, where the small-deflection beam assumption becomes less reliable.

3.2.2 Beam-model assessment under the 25 kg requirement

To assess whether the panel can support a 25 kg item, a maximum allowable bending angle of 3° (i.e., 0.052 rad) is assumed. The corresponding maximum deflection is

$$w_{max} = \frac{\theta L}{3} = 0.017\,L$$

Taking the span as L=1 m and the load as P=mg=245 N, the required critical bending stiffness is calculated as

$$E\,I_c = \frac{P\,L^3}{48\,w_{max}} = 300.25 \;\; N \cdot m^2$$

4. Discussion

4.1 Effect of layup on flexural stiffness and feasibility

The results of the three-point bending back-calculation indicate that the EI values of the three samples are ranked as follows: [0°/45°] < [0°/0°] < [0°/+45°/−45°/0°]. The four-layer sample has the highest EI. This is because the increase in thickness significantly enhances the moment of inertia of the cross-section, and the outer 0° layer is far from the neutral axis, which can more effectively bear the tensile and compressive stresses during the bending process. Although the ±45° layer in the middle is not the most efficient bearing layer for the main bending direction, it helps to improve the shear response and multi-directional stability.

The comparison of the two double-layer samples shows that under the current central bending load, arranging more 0° fibers along the main force direction is more conducive to improving the bending stiffness than introducing 45° fibers. Therefore, the EI and tensile modulus of [0°/0°] are higher than those of [0°/45°]. However, the tensile modulus can only reflect the axial stiffness in the plane, and the key issue for the aircraft stowage panel is the control of deflection under lateral loads. Therefore, the final evaluation should still be based on the EI obtained from the bending back-calculation.

Compared with the EI required under a 25 kg load, all three samples are significantly insufficient. Even though the four-layer sample performed the best in the experimental group, its EI is still far below the target value. Therefore, this lay-up scheme is only optimal within the experimental range and cannot be directly used for actual aircraft structural components.

4.2 A Rough Comparison with the Thickness of Real Aircraft Cargo Hold Panels

From a formula perspective, the thickness has a significant impact on the bending stiffness. For a rectangular cross-section

$$I = \frac{b\,h^3}{12},\; EI = E\,\frac{b\,h^3}{12}$$

Therefore, when the material modulus E and width b remain constant, EI is proportional to the cube of the thickness h. That is to say, if the thickness is doubled, the theoretical EI can increase by approximately 8 times; if the thickness is tripled, the theoretical EI can increase by approximately 27 times.

The actual aircraft interior panels are usually not simple hand-laid solid laminates like those in this experiment, but honeycomb sandwich composite structures. The Gellfab 5071A/5072A type sandwich panels used for aircraft cabin interior structures can be used in locations such as galley, partitions, storage compartments, etc., with a thickness range of 3.05–25.4 mm [3]. Additionally, in the research on aircraft storage bins, the test thickness of GFRP/Nomex sandwich panels is 15 mm, and the stowage bin demonstrators adopt a GFRP/Nomex honeycomb sandwich structure [4].

The thickness of the four-layer sample in this experiment is approximately 4.5 mm. If a rough estimation using the rectangular cross-section formula is made, the following table can be obtained.

Table 4-2-1. Simplified estimation of EI for representative thickness and size scenarios

Assumption	Thickness ratio vs. current sample	Estimated EI	Judgement
Current four-layer sample, 4.5 mm	1.00×	0.133 N·m²	Far below requirement
Same width, thickness increased to 15 mm [4]	3.33×	4.94 N·m²	Still far below
Width scaled to 1 m, thickness 15 mm [4]	3.33×	64.1 N·m²	Improved but insufficient
Width scaled to 1 m, thickness 25.4 mm [3]	5.64×	311 N·m²	Close to required EI

As shown in Table 4-2-1, if only the current four layers of samples are thickened to 15 mm, their estimated EI is still only 4.94 N·m², which is far below the required 300.25 N·m². Even if the plate width is further enlarged to 1 m according to the actual structural scale, the estimated EI at 15 mm thickness is only 64.1 N·m², still insufficient to meet the requirements. Only when the thickness is further increased to 25.4 mm, the estimated EI is close to the target value. This indicates that the gap between the current samples and the actual aircraft stowage panel is not only due to insufficient material thickness, but also due to the difference in structural form; the actual structure usually adopts a honeycomb sandwich design to more efficiently increase the cross-sectional moment of inertia and bending stiffness.

4.3 Limitations and Improvement Directions

The main limitations of this experiment stem from the hand layup process and the simplified three-point bending test. Due to the absence of vacuum bag pressure, rigid molds, or controlled compaction procedures, the samples have issues such as air bubbles, local pores, resin-rich areas, uneven resin distribution, thickness variations, aluminum foil adhesion, and residual edge tape. These defects weaken the load transfer between the glass fibers and the epoxy resin, preventing some fibers from fully bearing the load, thereby reducing the actual bending stiffness.

The test errors also affect the calculated results of EI. The dimensions and support spans of the three groups of samples were not the same. The mid-point deflection was manually read using a ruler, so the reading errors would affect the slope of the load-deflection curve. Moreover, only one sample was tested for each type of ply, lacking repetitive verification; the tensile test was only conducted on two double-layer samples, so it could only be used as an

auxiliary analysis.

Further improvements can be made in both the structural and manufacturing aspects. In terms of structure, the number of plies should be increased, the overall thickness of the sample should be enhanced, and a sandwich structure should be adopted. Also, the 0° main bearing layer should be placed near the outer surface to enhance the bending stiffness; the ±45° layers can be retained in the middle area to improve the shear response and multi-directional stability. In terms of manufacturing, rolling, press plates, or vacuum bag pressing should be used to reduce voids and resin-rich areas, and to improve fiber impregnation and resin/fiber ratio control.

5.Conclusion

In this experiment, three glass fiber/epoxy resin composite laminates were fabricated by hand lay-up method, and their mechanical properties were compared through three-point bending test and tensile test. The load-deflection data were used to calculate the bending stiffness EI based on the three-point bending model. The results showed that both the number of plies and the fiber orientation would affect the bending performance. Among them, the sample with four layers [0°/+45°/−45°/0°] had the highest EI, followed by the double-layer [0°/0°], and the lowest was the double-layer [0°/45°]. This indicates that increasing the thickness and arranging the 0° layer near the outer surface can effectively improve the bending stiffness.

However, compared with the EI required for the simplified 25 kg aircraft stowage-panel condition, all three samples were clearly insufficient. Although the four-layer mixed layup was the best solution in this experiment, it still could not directly meet the structural requirements of the aircraft luggage rack panel. Therefore, this experiment mainly demonstrated the influence of the layup design on the bending stiffness, and the current samples can only be used as laboratory-scale comparison samples.

In the future, the bending stiffness and stiffness-to-weight efficiency of composite materials should be enhanced by increasing the number of layers, sample thickness, optimizing the resin/fiber ratio, reducing voids and resin-rich areas, adopting pressure or vacuum bagging processes, and further considering the design of the core structure.

6. References

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7. AI Usage Statement

Figure 2-2-1 in this report was generated with the assistance of ChatGPT-5.4. The figure illustrates the specimen preparation process for cutting a rectangular composite ply while maintaining fibre geometry and preventing edge fraying. The figure shows a three-step workflow including preparation and marking, cutting along the taped region, and removing the finished ply with tape retained on the cut edges. The prompt used to generate this figure is as follows:
"Create a clean scientific infographic showing a three-step woven glass fibre cloth cutting procedure for composite material ply preparation. The image should be in a horizontal layout with three numbered panels connected by arrows. Use black, grey, and white technical drawing style with clear line art, subtle grid texture to represent woven glass fibre cloth, and minimal color. Panel 1: 'Prepare and mark'. Show a square woven glass fibre cloth sheet with a marked rectangular ply region in the center. Add paper tape around the marked rectangle before cutting. Include labels such as 'woven glass fibre cloth', 'marked rectangular ply region', and 'paper tape applied prior to cutting'. Panel 2: 'Cut along taped region'. Show scissors cutting along the taped outline. Indicate that the tape stabilizes the weave and prevents fibre fraying. Include labels such as 'scissors used to cut along taped region' and 'tape stabilizes the weave and prevents fibre fraying'. Panel 3: 'Remove ply with tape retained'. Show the finished rectangular ply separated from the remaining cloth, with tape retained on the cut edges. Include labels such as 'finished rectangular ply', 'tape retained on cut edges', and 'retained tape helps preserve ply geometry'. Use a clear academic technical illustration style suitable for a composite materials laboratory report."
Figure 2-2-1 is used solely as an illustrative diagram to support the experimental procedure, with the aim of enhancing the clarity and readability of the specimen cutting process described in this report. The content of this figure has been designed, selected, and verified based on my understanding of the experimental process, the underlying principles, and the requirements of this report. I have reviewed the experimental steps, textual descriptions, and technical implications presented in the figure and confirm that they correspond to the actual content of this experimental report.

Figure 2-2-2 in this report was generated with the assistance of ChatGPT-5.4. The figure illustrates the hand wet layup procedure for preparing composite laminate panels. The figure presents a six-step workflow, including placing the first ply on aluminium foil, mixing and applying resin, laying the next ply, repeating the layup sequence, applying the final top coat, and curing the laminate at room temperature. The prompt used to generate this figure is as follows:
"Create a clean scientific infographic titled 'Layup Procedure for Hand Wet Layup Panels'. The image should have six numbered panels arranged in a 3-by-2 grid with arrows showing the process flow. Use a technical line illustration style with black outlines, white background, light grey panels, pale yellow resin, and subtle woven texture for glass fibre plies. Panel 1: 'Place first ply on aluminium foil'. Show a pre-cut glass fibre ply placed on an aluminium foil substrate or liner. Include labels such as 'glass fibre ply' and 'aluminium foil substrate / liner'. Panel 2: 'Mix and apply resin'. Show resin part A and hardener part B mixed in a beaker using a stirrer, and a brush applying mixed resin onto the first ply. Include labels such as 'resin', 'hardener', 'mixed resin', and 'brush evenly wet first ply'. Panel 3: 'Lay next ply'. Show the next glass fibre ply being placed on top of the wetted ply, with a brush applying resin evenly. Add a note indicating that the next ply should follow the stacking sequence. Panel 4: 'Repeat layup sequence'. Show repeated placement of glass fibre plies and resin coats, with a small laminate cross-section illustrating alternating layers of glass fibre ply and resin-rich coat. Panel 5: 'Apply final top coat'. Show a brush applying the final resin top coat to seal the upper surface of the laminate. Panel 6: 'Cure'. Show the completed laminate panel curing at room temperature for approximately 24 hours, with a small clock symbol. Add a legend at the bottom for aluminium foil substrate, glass fibre ply, and mixed resin. Use clear labels and an academic laboratory-report style."
Figure 2-2-2 is used solely as an illustrative diagram to support the experimental procedure, with the aim of enhancing the clarity and readability of the hand wet layup process described in this report. The content of this figure has been designed, selected, and verified based on my understanding of the experimental process, the underlying principles, and the requirements of this report. I have reviewed the experimental steps, textual descriptions, and technical implications presented in the figure and confirm that they correspond to the actual content of this experimental report.

8. Contribution

All members participated in every group discussion session; the contribution chart does not accurately reflect each person's contributions, and in subsequent phases, everyone participated in revising other sections.

Members	LAB 1	LAB 2
Wang Yuhan	Overall experimental concept and steps summary Responsible for designing and manufacturing the samples Responsible for writing the materials selection, the design and production of samples section	Overall experimental concept and steps summary Responsible for preparing the samples of the fourth layer and recording experimental data Responsible for the writing of the "Layering Strategy and Design Basis" section
Zheng Junhao	Overall experimental concept and steps summary Responsible for the bending test Responsible for writing the experimental procedures and discussion section	Overall experimental concept and steps summary Responsible for preparing the four-layer samples and the bending test Responsible for writing the experimental procedures
Che Xuanming	Provide ideas for sample design Responsible for designing and manufacturing the samples Responsible for writing the discussion section	Provide ideas for sample design Responsible for preparing two parallel samples Responsible for writing the discussion section
Zhang Yuchen	Design bending test Responsible for conducting the bending test Responsible for writing the result section	Design bending test Responsible for preparing two parallel samples Responsible for writing the analysis section
Han Yujia	Provide ideas for sample design Responsible for recording experimental data Responsible for writing the conclusion section	Provide ideas for sample design Responsible for preparing the two-layer (0°, 45°) samples Responsible for writing the result section
Wang Yixuan	Design bending test Responsible for recording experimental data Responsible for writing for the introduction section	Design bending test Responsible for preparing the two-layer (0°, 45°) samples and recording experimental data Responsible for the introduction